Topic XI: Support Material

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APPENDIX A
Conversion Factors

multiply	by	to obtain
acres	0.4047	hectares
	43,560.0	square feet
	1.5625×10^{-3}	square miles
ampere-hours	3600.0	coulombs
angstrom units	3.937×10^{-9}	inches
	1×10^{-4}	microns
astronomical units	1.496×10^{8}	kilometers
atmospheres	76.0	centimeters of mercury
atomic mass unit	9.316×10^{8}	electron-volts
	1.492×10^{-10}	joules
	1.66×10^{-27}	kilograms
BeV (also GeV)	1×10^{9}	electron-volts
Btu	3.93×10^{-4}	horsepower-hours
	778.3	foot-pounds
	2.93×10^{-4}	kilowatt hours
	1.0×10^{-5}	therms
Btu/hr	0.2161	foot-pounds/sec
	3.929×10^{-4}	horsepower
	0.293	watts
bushels	2150.4	cubic inches
calories, gram (mean)	3.9683×10^{-3}	Btu (mean)
centares	1.0	square meters
centimeters	1×10^{-5}	kilometers
	1×10^{-2}	meters
	10.0	millimeters
	3.281×10^{-2}	feet
	0.3937	inches
chains	792.0	inches
coulombs	1.036×10^{-5}	faradays
cubic centimeters	0.06102	cubic inches
	2.113×10^{-3}	pints (U.S. liquid)
cubic feet	0.02832	cubic meters
	7.4805	gallons
cubic feet/min	62.43	pounds H_2O/min
cubic feet/sec	448.831	gallons/min
	0.64632	millions of gallons per day
cubits	18.0	inches
days	86,400.0	seconds
degrees (angle)	1.745×10^{-2}	radians
degrees/sec	0.1667	revolutions/min
dynes	1×10^{-5}	newtons
electron-volts	1.074×10^{-9}	atomic mass units
	1×10^{-9}	BeV (also GeV)
	1.602×10^{-19}	joules
	1.783×10^{-36}	kilograms
	1×10^{-6}	MeV
faradays/sec	96,500	amperes (absolute)
fathoms	6.0	feet
feet (international)	30.48	centimeters
	0.3048	meters
feet (U.S. Survey)	1200/39.37	centimeters
	12/39.37	meters
	1.645×10^{-4}	miles (nautical)
	1.894×10^{-4}	miles (statute)

multiply	by	to obtain
feet/min	0.5080	centimeters/sec
feet/sec	0.592	knots
	0.6818	miles/hr
foot-pounds	1.285×10^{-3}	Btu
	5.051×10^{-7}	horsepower-hours
	3.766×10^{-7}	kilowatt-hours
foot-pound/sec	4.6272	Btu/hr
	1.818×10^{-3}	horsepower
	1.356×10^{-3}	kilowatts
furlongs	660.0	feet
	0.125	miles (statute)
gallons	0.1337	cubic feet
	3.785	liters
gallons H_2O	8.3453	pounds H_2O
gallons/min	8.0208	cubic feet/hr
	0.002228	cubic feet/sec
GeV (also BeV)	1×10^{9}	electron-volts
grams	1×10^{-3}	kilograms
	3.527×10^{-2}	ounces (avoirdupois)
	3.215×10^{-2}	ounces (troy)
	2.205×10^{-3}	pounds
hectares	2.471	acres
	1.076×10^{5}	square feet
horsepower	2545.0	Btu/hr
	42.44	Btu/min
	550	foot-pounds/sec
	0.7457	kilowatts
	745.7	watts
horsepower-hours	2545.0	Btu
	1.976×10^{-6}	foot-pounds
	0.7457	kilowatt-hours
hours	4.167×10^{-2}	days
	5.952×10^{-3}	weeks
inches	2.540	centimeters
	1.578×10^{-5}	miles
inches, H_2O	5.199	pounds force/ft^2
	0.0361	psi
	0.0735	inches, mercury
inches, mercury	70.7	pounds force/ft^2
	0.491	pounds force/in^2
	13.60	inches, H_2O
joules	6.705×10^{9}	atomic mass units
	9.480×10^{-4}	Btu
	1×10^{7}	ergs
	6.242×10^{18}	electron-volts
	1.113×10^{-17}	kilograms
kilograms	6.025×10^{26}	atomic mass units
	5.610×10^{35}	electron-volts
	8.987×10^{16}	joules
	2.205	pounds
kilometers	3281.0	feet
	1000.0	meters
	0.6214	miles
kilometers/hr	0.5396	knots

Support Material

APPENDIX A *(continued)*
Conversion Factors

multiply	by	to obtain
kilowatts	3412.9	Btu/hr
	737.6	foot-pounds/sec
	1.341	horsepower
kilowatt-hours	3413.0	Btu
knots	6076.0	feet/hr
	1.0	nautical miles/hr
	1.151	statute miles/hr
light years	5.9×10^{12}	miles
links (surveyor)	7.92	inches
liters	1000.0	cubic centimeters
	61.02	cubic inches
	0.2642	gallons (U.S.liquid)
	1000.0	milliliters
	2.113	pints
MeV	1×10^{6}	electron-volts
meters	100.0	centimeters
	1/0.3048	feet (international)
	39.37/12	feet (U.S. Survey)
	1×10^{-3}	kilometers
	5.396×10^{-4}	miles (nautical)
	6.214×10^{-4}	miles (statute)
	1000.0	millimeters
microns	1×10^{-6}	meters
miles (nautical)	6076	feet
	1.853	kilometers
	1.1516	miles (statute)
miles (statute)	5280.0	feet
	1.609	kilometers
	0.8684	miles (nautical)
miles/hr	88.0	feet/min
milligrams/liter	1.0	parts/million
milliliters	1×10^{-3}	liters
millimeters	3.937×10^{-2}	inches
newtons	1×10^{5}	dynes
ohms (international)	1.0005	ohms (absolute)
ounces	28.349527	grams
	6.25×10^{-2}	pounds
ounces (troy)	1.09714	ounces (avoirdupois)
parsecs	3.086×10^{13}	kilometers
	1.9×10^{13}	miles
pascal-sec	1000	centipoise
	10	poise
	0.02089	pound force-sec/ft^2
	0.6720	pound mass/ft-sec
	0.02089	slug/ft-sec

multiply	by	to obtain
pints (liquid)	473.2	cubic centimeters
	28.87	cubic inches
	0.125	gallons
	0.5	quarts (liquid)
poise	0.002089	pound-sec/ft^2
pounds	0.4536	kilograms
	16.0	ounces
	14.5833	ounces (troy)
	1.21528	pounds (troy)
pounds/ft^2	0.006944	pounds/in^2
pounds/in^2	2.308	feet, H_2O
	27.7	inches, H_2O
	2.037	inches, mercury
	144	pounds/ft^2
quarts (dry)	67.20	cubic inches
quarts (liquid)	57.75	cubic inches
	0.25	gallons
	0.9463	liters
radians	57.30	degrees
	3438.0	minutes
revolutions	360.0	degrees
revolutions/min	6.0	degrees/sec
rods	16.5	feet
	5.029	meters
rods (surveyor)	5.5	yards
seconds	1.667×10^{-2}	minutes
square meters/sec	1×10^{6}	centistokes
	10.76	square feet/sec
	1×10^{4}	stokes
slugs	32.174	pounds mass
stokes	0.0010764	square feet/sec
tons (long)	1016.0	kilograms
	2240.0	pounds
	1.120	tons (short)
tons (short)	907.1848	kilograms
	2000.0	pounds
	0.89287	tons (long)
volts (absolute)	3.336×10^{-3}	statvolts
watts	3.4129	Btu/hr
	1.341×10^{-3}	horsepower
yards	0.9144	meters
	4.934×10^{-4}	miles (nautical)
	5.682×10^{-4}	miles (statute)

APPENDIX B
Surveying Conversion Factors

multiply	by	to obtain
ac	43,560	ft^2
	10	$chains^2$
	4046.87	m^2
ac-ft	43,560	ft^3
	1233.49	m^3
chain	66	ft
	22	yd
	4	rods
day (mean solar)	86,400	sec
day (sidereal)	86,164.09	sec
deg (angle)	0.0174533	rad
	17.77778	mils
engineer's link	1	ft
ft (U.S.Survey)	0.3048006	m
grads	0.9	degrees (angle)
	0.01570797	rad
hectare	2.47104	ac
	10,000	m^2
in	25.4	mm
labors	177.14	ac
leagues	4428.40	ac
link (see engineer's link and surveyor's link)		
mils	0.05625	degrees (angle)
	3,037.500	min
mi (statute)	5280	ft
	80	chains (surveyor's)
	320	rods
	0.86839	mi (nautical)
mi^2	640	ac
	27,878,400	ft^2
min (angle)	0.29630	mils
	0.000290888	rad
min (mean solar)	60	sec
min (sidereal)	59.83617	sec
outs	330	ft
	10	33 ft chains
rad	57.2957795	degrees (angle)
	57°17′44.806″	degrees (angle)
rods	16.5	ft
	1	perches
	1	poles

Support Material

APPENDIX B *(continued)*
Surveying Conversion Factors

multiply	by	to obtain
sec (angle)	4.848137×10^{-6}	rad
sec (sidereal)	0.9972696	sec (mean solar)
surveyor's link	0.66	ft
	7.92	in
VARA (California)	33	in
VARA (Texas)	$33\frac{1}{3}$	in
yd (U.S.)	0.914402	m

APPENDIX C
Miscellaneous Constants and Conversions

0.0000001 per °F	= coefficient of expansion for invar tape
0.00000645 per °F	= coefficient of expansion for steel tape
0.6745	= coefficient for 50% standard deviation
1.15 mi	= 1 minute of latitude
1.6449	= coefficient for 90% standard deviation
6 mi	= length and width of township
10 square chains	= 1 ac
15° longitude	= width of one time zone
23°26.5′	= maximum declination of the sun at solstice
24 hr	= 360° of longitude
36	= number of sections in a township
69.1 mi	= 1° latitude
100	= usual stadia ratio
101 ft	= 1 second of latitude
400 grads	= 360°
480 chains	= width and length of township
640 ac	= 1 normal section
4046.9 mi^2	= 1 ac
6400 mils	= 360°
43,560 ft^2	= 1 ac
20,906,000 ft	= mean radius of earth

APPENDIX D
Glossary

Not all words are covered in this glossary.
Refer to the index for additional words that are discussed in the text.

Abstract – A summary of facts.

Abstract of title – A condensed history of the title to land.

Accessory to corner – A physical object that is adjacent to a corner. An accessory is usually considered part of the monument.

Acclivity – An upward slope of ground.

Accretion – The gradual accumulation of land by natural causes.

Acknowledgment – A declaration by a person before an official (usually a notary public) that he or she executed a legal document.

Acquiescence – Implied consent to a transaction, to the accrual of a right, or to any act, by one's silence (or without express assent).

Adjudication – The giving or pronouncing of a judgment or decree.

Adverse possession – A method of acquiring property by holding it for a period of time under certain conditions.

Affidavit – A written declaration under oath before an authorized official (usually a notary public).

Alienation – The transfer of property and/or possessions from one person to another.

Aliquot – A portion contained in something else a whole number of times.

Alluvium – Sand or soil deposited by streams.

Appellant – The party that takes an appeal from one court or jurisdiction to another.

Appurtenance – A right, privilege, or improvement belonging to and passing with a piece of property when it is conveyed.

Assigns – Those to whom property is transferred.

Avulsion – A sudden and perceptible change of shoreline by the violent action of water.

Bayou – An outlet from a swamp or lagoon to the sea.

Bed of stream – The depression between the banks of a water course worn by the regular and usual flow of the water.

Bequest – A gift by will of personal property.

Bounty lands – Portions of the public domain given or donated as a bounty for services rendered.

Chain of title – A chronological list of documents that comprise the record history of title of real property.

Civil law – That part of the law pertaining to civil rights, as distinguished from criminal law. Civil law and Roman law have the same meaning. In contradistinction to English common law, civil law is enacted by legislative bodies.

Clear title – Good title. Title free from encumbrances.

Cloud on title – A claim or encumbrance on a title to land that may or may not be valid.

Color of title – Any written instrument that appears to convey title, even though it does not.

Common law – Principles and rules of action determined by court decisions that have been accepted by generation after generation, and that are distinguished from laws enacted by legislative bodies.

Consideration – Something of value given to make an agreement binding.

Conveyance – Any instrument in writing by which an interest in real property is transferred.

Covenant – When used in deeds, restrictions imposed on the grantee as to the use of land conveyed.

Crown – The sovereign power in a monarchy.

Cut bank – The watershed and relatively permanent elevation or acclivity that separates the bed of a river from its adjacent upland.

Decree – A judgment by the court in a legal proceeding.

Dedication – An appropriation of land to some public use made by the owner, and accepted for such use by or on behalf of the public.

Deed – Evidence in writing of the transfer of real property.

Deed of trust – An instrument taking the place of a mortgage, by which the legal title to real property is placed in one or more trustees to secure repayment of a sum of money.

Demurrer – In legal pleading, the formal mode of disputing the sufficiency of the pleading of the other side.

Devise – A gift of real property by the last will and testament of the donor.

Easement – The right that the public, an individual, or individuals have in the lands of another.

Egress – The right or permission to go out from a place; right of exit.

APPENDIX D *(continued)*
Glossary

Eminent domain – The right or power of government or certain other agencies to take private property for public use on payment of just compensation to the owner.

Encroachment – An obstruction that intrudes upon the land of another. The gradual, stealthy, illegal acquisition of property.

Encumbrance – Any burden or claim on property, such as a mortgage or delinquent taxes.

Equity – The excess of the market value over any indebtedness.

Erosion – The process by which the surface of the earth is worn away by the action of waters, glaciers, wind, or waves.

Escheat – Reversion of property to the state where there is no competent or available person to inherit it.

Escrow – Something placed in the keeping of a third person for delivery to a given party upon fulfillment of some condition.

Estate – An interest in property, real or personal.

Estoppel – A bar or impediment that precludes allegation or denial of a certain fact or state of facts in consequence of a final adjudication.

Et al. – An abbreviation for "and others."

Et Mode Ad Hune Diem – An abbreviation for "and now at this day."

Et ux – An abbreviation for "and wife."

Evidence aliunde – Evidence from outside or from another source.

Extrinsic evidence – Evidence NOT contained in the deed, but offered to clear up an ambiguity found to exist when applying the description to the ground.

Grant – A transfer of property.

Grantee – The person to whom a grant is made.

Grantor – The person by whom a grant is made.

Good faith – An honest intention to abstain from taking advantage of another.

Gradient – An inclined surface. The change in elevation per unit of horizontal distance.

Hereditament – Something capable of being inherited, be it real or personal property.

Hiatus – An area between two surveys of record described as having one or more common boundary lines with no omission.

Holograph – A will written entirely by the testator in his or her own handwriting.

Incumbrance – A right, interest in, or legal liability upon real property that does not prohibit passing title to the land but that diminishes its value.

Ingress – The right or permission to go upon a place; right of entrance.

Intent – The true meaning (from the written words of an instrument).

Intestate – Without making a will.

Judgment – The official and authentic decision of a court of justice.

Leasehold – An estate in realty held under a lease; an estate for a fixed term of years.

Lessee – The person to whom a lease is made.

Lessor – The person who grants a lease.

Lien – A claim or charge on property for payment of some debt, obligation, or duty.

Lis pendens – A pending suit. A notice of lis pendens is filed for the purpose of warning all persons that a suit is pending.

Litigation – Contest in a court of justice for the purpose of enforcing a right.

Littoral – Belonging to the shore, as of seas and lakes.

Logical relevancy – A relationship in logic between the fact for which evidence is offered and a fact in issue such that the existence of the former renders probable or improbable the existence of the latter.

Mean – Intermediate; the middle between two extremes.

Memorial – That which contains the particulars of a deed, and so on. In practice, a memorial is a short note, abstract, memorandum, or rough draft of the orders of the court, from which the records thereof may at any time be fully made up.

Mortgage – A conditional conveyance of an estate as a pledge for the security of a debt.

Muniment – Documentary evidence of title.

Option – The right as granted in a contract or by an initial payment of acquiring something in the future.

Parcel – A part of a piece of land that cannot be identified by a lot or tract number.

Parol evidence – Evidence that is given verbally.

Patent – A government grant of land. The instrument by which a government conveys title to land.

APPENDIX D *(continued)*
Glossary

Plat – A scaled diagram showing boundaries of a tract of land or subdivisions. May constitute a legal description of the land and be used in lieu of a written description.

Power of attorney – A written document given by one person to another authorizing the latter to act for the former.

Prescription – Creation of an easement under claim of right by use of land that has been open, continuous, and exclusive for a period of time prescribed by law.

Prima facie evidence – Facts presumed to be true unless disproved by evidence to the contrary.

Privity – The relationship that exists between parties to a contract. Mutual or successive relationship to the same rights of property, such as the relationship of heir with ancestor or donee with donor.

Privy – A person who is in privity with another.

Probate – The act or process of validating a will.

Quiet title – Action of law to remove an adverse claim or cloud on title.

Quitclaim deed – A conveyance that passes any title, interest, or claim that the grantor may have.

Reliction – A gradual and imperceptible recession of water, resulting in increased shoreline, beach, or property.

Relinquishment – The forsaking, abandonment, renouncement, or gift of a right.

Remand – To send a cause back to the same court out of which it came for the purpose of having some action taken upon it there.

Riparian – Belonging or relating to the bank of a river.

Royalty – A share of the profit from sale of minerals paid to the owner of the property by the lessee.

Said – Refers to one previously mentioned.

Scrivener – A person whose occupation is to draw up contracts, write deeds and mortgages, and prepare other written instruments.

Shore – The place lying between the line of ordinary high tide and the line of lowest tide.

Sovereign – A person, body, or state in which independent and supreme authority is vested.

Squatter – One who settles on another's land without legal authority.

Statute – A particular law established by the legislative branch of government.

Statutory – Relating to a statute.

Submerged land – In tidal areas, land that extends seaward from the shore and is continuously covered during the ebb and flow of the tide.

Substantive evidence – Evidence used to prove a fact (as opposed to evidence given for the purpose of discrediting a claim).

Tenancy by the entirety – Husband and wife each possesses the entire estate in order that, upon the death of either spouse, the survivor is entitled to the estate in its entirety.

Tenancy, Joint – The holding of property by two or more persons, each of whom has an undivided interest. After the death of one of the joint tenants, the surviving tenant(s) receive the descendent's share.

Tenant – One who has the temporary use and occupation of real property owned by another person (the landlord).

Tenements – Property held by a tenant. Everything of a permanent nature. In a more restrictive sense, a house or dwelling.

Testament – A will of personal property.

Testator – One who makes a testament or will. One who dies leaving a will.

Thalweg – The deepest part of a channel.

Thence – From that place; the following course is continuous from the one before it.

Title policy – Insurance against loss or damage resulting from defects or failure of title to a particular parcel of land.

To wit – That is to say; namely.

Upland – Land above mean high water and subject to private ownership (as distinguished from tidelands, which are in the state). Also used as meaning nonriparian.

Watercourse – A running stream of water fed from permanent or natural sources running in a particular direction and having a channel formed by a well-defined bed and banks (though it need not flow continuously).

Warranty deed – A deed in which the grantor proclaims that he or she is the lawful owner of real property and will forever defend the grantee against any claim on the property.

Will – The legal declaration of a person's wishes as to the disposition of his or her property after his or her death.

APPENDIX D *(continued)*
Glossary

Witness mark – A mark placed at a known location to aid in recovery and identification of a monument or corner.

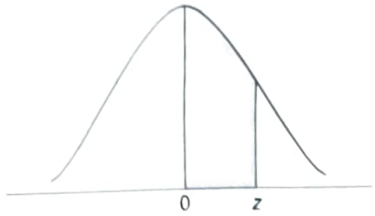
Writ – A mandatory order issued from a court of justice.

Writ of coram nobis – A common law writ, the purpose of which is to correct an error in a judgment in the same court in which it was rendered.

APPENDIX E

Areas Under the Standard Normal Curve

(0 to z)



z	0	1	2	3	4	5	6	7	8	9
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0754
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2258	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2518	0.2549
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2996	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

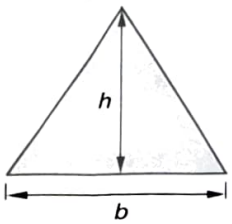
APPENDIX F
Mensuration of Two-Dimensional Areas

Nomenclature

A	total surface area
b	base
c	chord length
d	distance
h	height
L	length
p	perimeter
r	radius
s	side (edge) length, arc length
θ	vertex angle, in radians
ϕ	central angle, in radians

Triangle

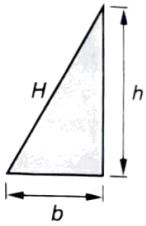
equilateral



$$A = \tfrac{1}{2}bh = \frac{\sqrt{3}}{4}b^2$$

$$h = \frac{\sqrt{3}}{2}b$$

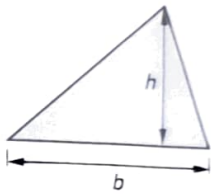
right



$$A = \tfrac{1}{2}bh$$

$$H^2 = b^2 + h^2$$

oblique



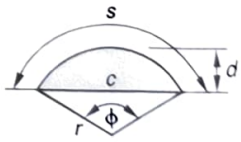
$$A = \tfrac{1}{2}bh$$

Circle

$$p = 2\pi r$$

$$A = \pi r^2 = \frac{p^2}{4\pi}$$

Circular Segment

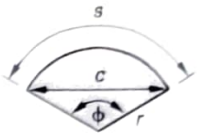


$$A = \tfrac{1}{2}r^2(\phi - \sin\phi)$$

$$\phi = \frac{s}{r} = 2\left(\arccos\frac{r-d}{r}\right)$$

$$c = 2r\sin\left(\frac{\phi}{2}\right)$$

Circular Sector



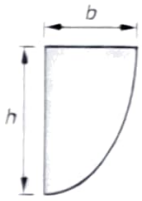
$$A = \tfrac{1}{2}\phi r^2 = \tfrac{1}{2}sr$$

$$\phi = \frac{s}{r}$$

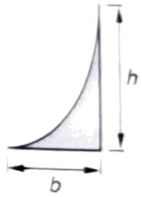
$$s = r\phi$$

$$c = 2r\sin\left(\frac{\phi}{2}\right)$$

Parabola

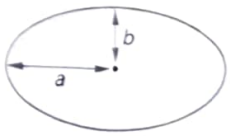


$$A = \tfrac{2}{3}bh$$



$$A = \tfrac{1}{3}bh$$

Ellipse



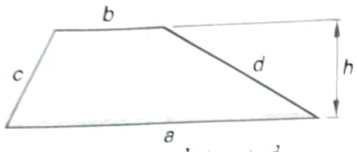
$$A = \pi ab$$

$$p \approx 2\pi\sqrt{\tfrac{1}{2}(a^2 + b^2)} \quad \left[\text{Euler's upper bound}\right]$$

Support Material

APPENDIX F *(continued)*
Mensuration of Two-Dimensional Areas

Trapezoid

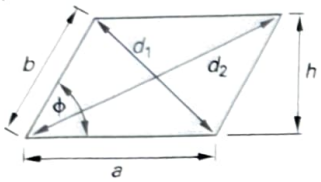


$$p = a + b + c + d$$

$$A = \tfrac{1}{2}h(a + b)$$

If $c = d$, the trapezoid is isosceles.

Parallelogram



$$p = 2(a + b)$$

$$d_1 = \sqrt{a^2 + b^2 - 2ab\cos\phi}$$

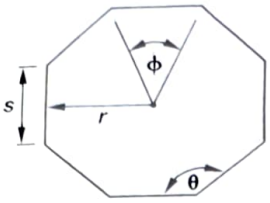
$$d_2 = \sqrt{a^2 + b^2 + 2ab\cos\phi}$$

$$d_1^2 + d_2^2 = 2(a^2 + b^2)$$

$$A = ah = ab\sin\phi$$

If $a = b$, the parallelogram is a rhombus.

Regular Polygon (*n* equal sides)



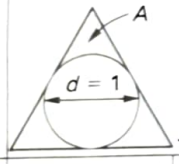
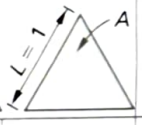
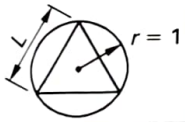
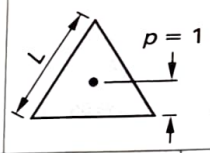
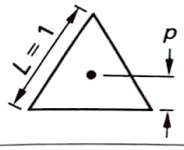
$$\phi = \frac{2\pi}{n}$$

$$\theta = \frac{\pi(n-2)}{n} = \pi - \phi$$

$$p = ns$$

$$s = 2r\tan\frac{\theta}{2}$$

$$A = \tfrac{1}{2}nsr$$

regular polygons							
sides	name	area (A) when diameter of inscribed circle = 1	area (A) when side = 1	radius (r) of circumscribed circle when side = 1	length (L) of side when radius (r) of circumscribed circle = 1	length (L) of side when perpendicular to circle = 1	perpendicular (p) to center when side = 1
3	triangle	1.299	0.433	0.577	1.732	3.464	0.289
4	square	1.000	1.000	0.707	1.414	2.000	0.500
5	pentagon	0.908	1.720	0.851	1.176	1.453	0.688
6	hexagon	0.866	2.598	1.000	1.000	1.155	0.866
7	heptagon	0.843	3.634	1.152	0.868	0.963	1.038
8	octagon	0.828	4.828	1.307	0.765	0.828	1.207
9	nonagon	0.819	6.182	1.462	0.684	0.728	1.374
10	decagon	0.812	7.694	1.618	0.618	0.650	1.539
11	undecagon	0.807	9.366	1.775	0.563	0.587	1.703
12	dodecagon	0.804	11.196	1.932	0.518	0.536	1.866

APPENDIX G
Mensuration of Three-Dimensional Volumes

Nomenclature

A	surface area
b	base
h	height
r	radius
R	radius
s	side (edge) length
V	internal volume

Sphere

$$V = \tfrac{4}{3}\pi r^3 = \tfrac{4}{3}\pi\left(\frac{d}{2}\right)^3 = \tfrac{1}{6}\pi d^3$$

$$A = 4\pi r^2$$

Right Circular Cone (excluding base area)

$$V = \tfrac{1}{3}\pi r^2 h = \tfrac{1}{3}\pi\left(\frac{d}{2}\right)^2 h = \tfrac{1}{12}\pi d^2 h$$

$$A = \pi r\sqrt{r^2 + h^2}$$

Right Circular Cylinder (excluding end areas)

$$V = \pi r^2 h$$

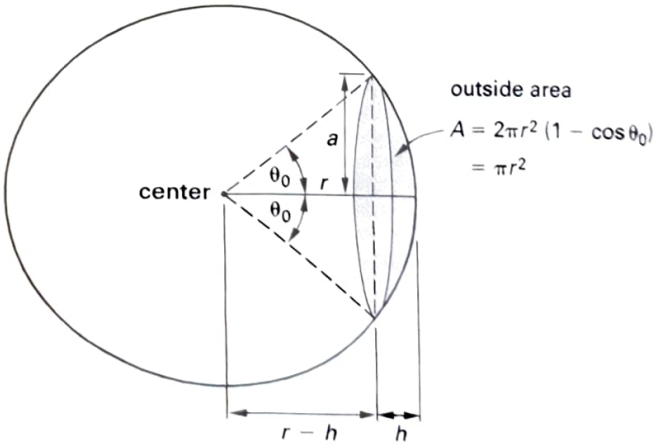
$$A = 2\pi r h$$

Spherical Segment (spherical cap)

Surface area of a spherical segment of radius r cut out by an angle θ_0 rotated from the center about a radius, r, is

$$A = 2\pi r^2(1 - \cos\theta_0)$$

$$\omega = \frac{A}{r^2} = 2\pi(1 - \cos\theta_0)$$

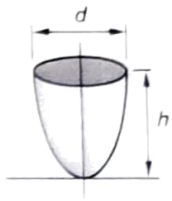


$$V_{\text{cap}} = \tfrac{1}{6}\pi h(3a^2 + h^2)$$

$$= \tfrac{1}{3}\pi h^2(3r - h)$$

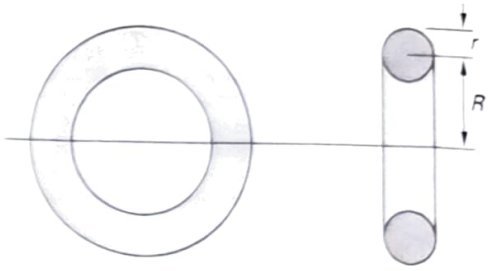
$$a = \sqrt{h(2r - h)}$$

Paraboloid of Revolution



$$V = \tfrac{1}{8}\pi d^2 h$$

Torus



$$A = 4\pi^2 Rr$$

$$V = 2\pi^2 Rr^2$$

Regular Polyhedra (identical faces)

name	number of faces	form of faces	total surface area	volume
tetrahedron	4	equilateral triangle	$1.7321s^2$	$0.1179s^3$
cube	6	square	$6.0000s^2$	$1.0000s^3$
octahedron	8	equilateral triangle	$3.4641s^2$	$0.4714s^3$
dodecahedron	12	regular pentagon	$20.6457s^2$	$7.6631s^3$
icosahedron	20	equilateral triangle	$8.6603s^2$	$2.1817s^3$

The radius of a sphere inscribed within a regular polyhedron is

$$r = \frac{3V_{\text{polyhedron}}}{A_{\text{polyhedron}}}$$

Support Material